

Common Laboratory Equipment

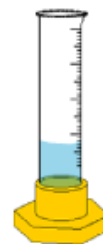
Beaker



Erlenmeyer flask



scoopula



graduated cylinder



test tube



test tube holder



crucible



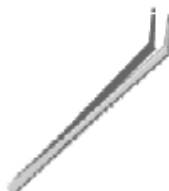
crucible tongs



stand and ring clamp



clay triangle



forceps



wire screen



dropper pipet



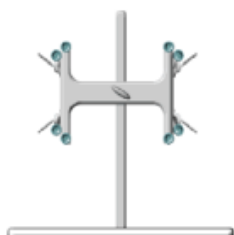
utility clamp



pinch clamp



funnel



buret clamp



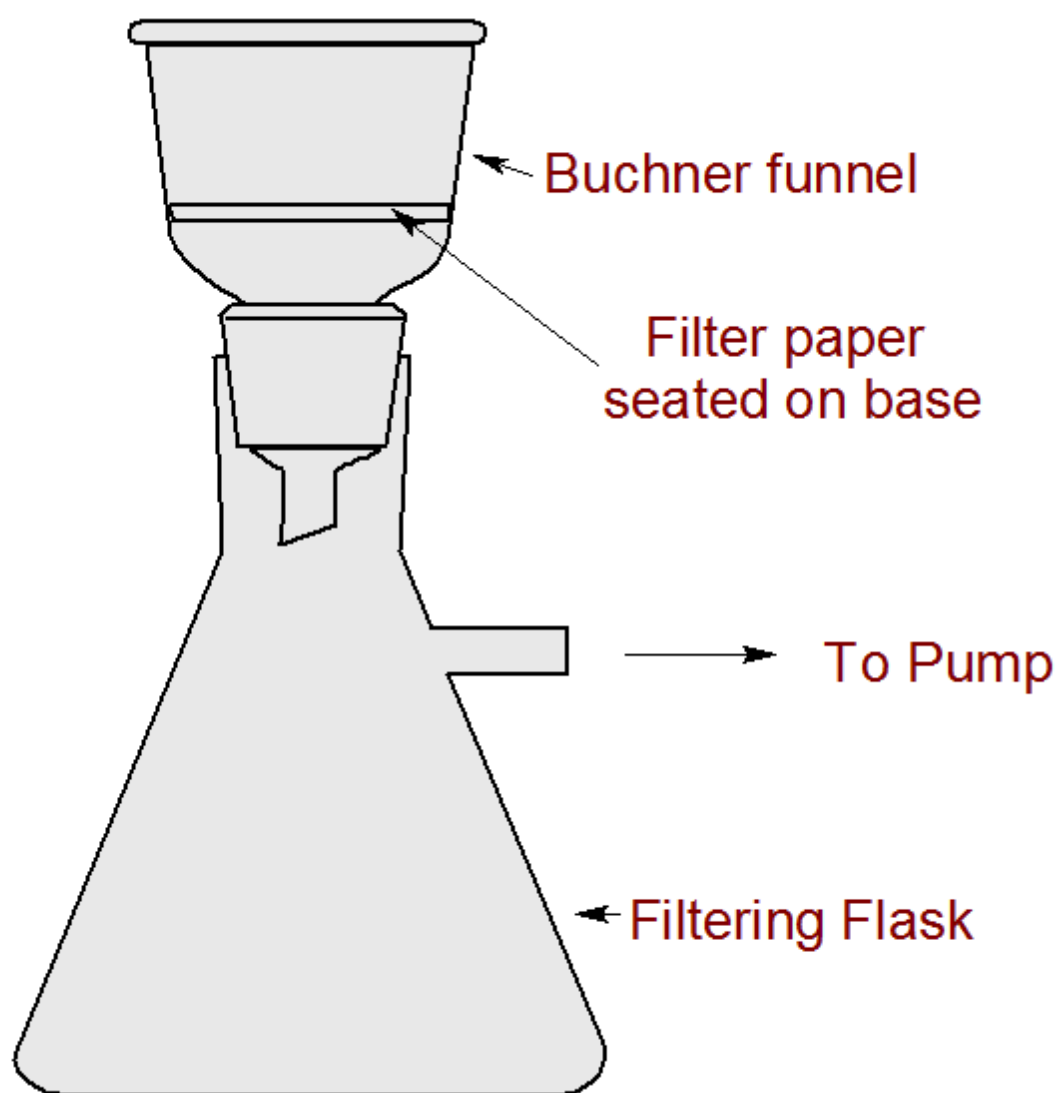
evaporating dish



watch glass



rubber policeman



Memorize the parts

Laboratory Safety Rules:

- 1) wear goggles(EYE PROTECTION), lab coat.
- 2) No eating/drinking or smoking.
- 3) No accompanying friends or children.
- 4) No direct smell of evolving gases(but by fanning).
- 5) Never return the non used substance to the original bottle.
- 6) We always add the acid into the water H_2SO_4 into H_2O .
- 7) All experiments done must be approved by the instructor.
- 8) Attending late to the lab isn't allowed.
- 9) Your shoes must cover the entire foot and you must not put lenses.
- 10) Girls must tie their hair and never keep it loose.
- 11) If anything wrong happens; such as cutting yourself or any accident, you must go and tell your instructor about it.

Most important conversions:

- 1) 1 nanometer (nm) = 10^{-9} meter
- 2) 1 Angstrom (A) = 10^{-10} meter
- 3) 1 micrometer (μm) = 10^{-6} meter
- 4) 1 millimeter (mm) = 10^{-3} meter
- 5) 1 centimeter (cm) = 10^{-2} meter
- 6) 1 decimeter (dm) = 10^{-1} meter
- 7) 1 liter (L) = 1 dm^3
- 8) $1 \text{ cm}^3 = 1 \text{ mL}$
- 9) 1000 gram = 1 kilo gram

Temperature unit conversion:

$$* \quad K = C + 273$$

$$* \quad F = \frac{9}{5}C + 32$$

K = KELVIN

C = CELSIUS

F = Fahrenheit

QUESTIONS

True or False : example (1) : unused reagents must be returned to the original bottle.

. example (2) : it is unwise to mix household chemicals .

Answer: (1) false / (2) true

EXPERIMENT (1)

DENSITY AND LABORATORY TECH.

Objective (الهدف من التجربة): determine the density of an unknown liquid and an irregular solid at a given temperature.

الاسئلة بالامتحان ستكون معظمها من هدف التجارب

Example: what is the density for

$$\text{DENSITY} = \frac{\text{mass}(g)}{\text{volume}(mL)} , 1\text{mL}=1\text{cm}^3$$

- Density is temperature “dependent”
- Density is an intensive property (independent on the amount (mass,volume) of the substance.
- Density is not an “extensive” property
- Density inversely proportional to the temperature

$$\text{Density} \uparrow = \frac{\text{mass}}{\text{volume}(\text{temperature})} \downarrow$$

الكثافة تتناسب تناسبا عكسيا
مع الحجم (الحرارة) .
بحيث كل ما زادت الحرارة
قلت الكثافة.

Precision of balance depends on number of decimal points <examples below>

0.00 → precision is ± 0.01
0.000 → precision is ± 0.001
0.0000 → precision is ± 0.0001

Example : what is the precision of an object that has a mass of 6.732grams ?

Answer : ± 0.001

- Precision of cylinder equal to the half of the lowest scaling . so if the lowest scaling equals to 0.1cm³ so the precision = 0.1/2 = ±0.05

Significant Figures : we use sig.fig to show more certainty, so as the number of sig.fig increases the more certain the measurement.

Counting RULES:

1) ALL non zero numbers are always significant (1,2,3,4,5,6,7,8,9)

example: 21 → 2 sig.fig

2) Zeros between nonzero digits are always significant

example: 123001 → 6 sig.fig

3) All zeros to the left of the first non-zero (1,2,3,4,5,6,7,8,9) are NOT significant.

Example: 0.0505 → 3 sig.fig

example: 0.00000005 → 1 sig.fig

4) If there is a decimal zero to the right of the last non-zero are always significant.

Example: 2.0 → 2 sig.fig

Examples of Significant Figures

- **Determine the number of significant figures in the following measurements:**

- | | |
|----------------------------------|------------------------|
| 1. 328 cm | Answer: 3 |
| 2. 6.01 g | Answer : 3 |
| 3. 0.805 m | Answer : 3 |
| 4. 0.043 kg | Answer : 2 |
| 5. 1.510×10^{22} atoms | Answer : 4 |
| 6. 5000 mL | Answer : 1, 2, 3, or 4 |
| 7. 5.0×10^3 mL | Answer : 2 |
| 8. 5.000×10^3 mL | Answer : 4 |
| 9. 0.030200 m ³ | Ans: 5 |
| 10. 6.40×10^4 molecules | Ans: 3 |

- **ADDING AND SUBTRACTING** : count the number of decimal places. Round to the lowest number of decimal places. And if you are trying to remove a figure from an answer round up if 5 or higher.

Examples :

$$2.\underline{5} + 2 = 4.5 \text{ which round to } 5$$

There is only one decimal place in 2.5/ there isn't any decimal place in 2 so it counts as zero decimal so we compare between one and zero then the answer must be least

$$5.\underline{31} - 2.\underline{2} = 3.11 \text{ which rounds to } 3.1$$

There are 2 decimal places in 5.31 and there is only 1 in 2.2 so the answer must contain the least which is 1 decimal point only

- **MULTIPLYING AND DIVIDING**: count the number of significant figures. Round to the lowest number of significant. Remember, if you are trying to remove a figure round if it is 5 or higher.

example → $2.55 * 2 = 5.1$ which rounds to 5

in MULTIPLYING AND DIVIDING
we count how many sig.fig there
is

في هاذ المثال "سواء كان ضرب او قسمة" بنعد اكم
سيجنيفكانت فيجر بكل عدد و الجواب النهائي لازم
يكون الاقل بينهما . يعني (2.55 فيها 3) (2 فيها 1)
فالجواب النهائي لازم يحتوي واحد فقط

more examples



$$10.0 / 3.0 = 3.333333 \text{ which rounds to } 3.3$$

$$16.5 / 3 = 5.5 \text{ which rounds to } 6$$

- In ADDING AND SUBTRACTING the last answer should have least decimals
- In MULTIPLYING AND DIVIDING the last answer is with least sig.fig

Accuracy: how a measured values are close to the actual value.

Precision: how a measured values are close to each other.

PRECISION VS ACCURACY



✓ Precision
✗ Accuracy



✗ Precision
✓ Accuracy



✗ Precision
✗ Accuracy



✓ Precision
✓ Accuracy

QUESTIONS

1) When 4.35g of iron immersed in 3.35mL of water, the volume was increased to 3.89mL calculate the density of iron ? (consider significant figures)

Solution: $\text{Density} = \frac{\text{mass}}{\text{volume}} = \text{Density} = \frac{\text{mass}}{\Delta \text{volume}} = \frac{4.35}{3.89 - 3.35} = 8.05555 = 8.1$

2) A 56.0 g liquid sample has a density of 0.880 g/mL the volume of this sample in mL is ? (use sig.fig)

A) 63.6

B) 85.4

C) 75

D) 68

Answer : A

3) give the answer the proper number of significant figures: $(3.50 + 0.10)(3.05 - 0.10)$?

A) 10.6

B) 0.65

C) 6.55

D) 1.22

Answer : A

4) in SI units, what is the unit of mass, volume and density ?

Answer: respectively : $\text{Kg} / \text{m}^3 / \frac{\text{Kg}}{\text{m}^3}$

(س3) بنحسب كل عملية لحال بدون
القواعد بعددين بنستخدم قاعدة
الضرب بالنهاية
(يعني بالخطوة الاخيرة فقط)

5) true or false : density of liquids depend on temperature ? True

EXPERIMENT (2)

PHYSICAL SEPARATION OF MIXTURES.

Objective: Separate the following mixture $\text{NH}_4\text{Cl}/\text{NaCl}/\text{SiO}_2$ and determine the percentage % composition.

A Mixture: is a physical combination between two or more components in which each component keeps its own identity(properties).

TYPES

- Homogeneous(متجانس): any solution/air/ $\text{NaCl}/\text{NH}_4\text{Cl}$
- Heterogeneous(غير متجانس): two or more separate phases like;($\text{NaCl}, \text{SiO}_2$) ($\text{NH}_4\text{Cl}, \text{SiO}_2$)

Methods of Separation:

Vaporizing : تبخير
condensing : التكثيف
Solvent : المذيب

- **Sublimation**(التسامي): converting solid to gas, vaporizing a solid and condensing its vapor and not all solids sublime.

Example: NH_4Cl

- **Extraction**(الاستخلاص): pulling out a desired substance from one phase by another by adding a solvent.

Example: NaCl

- **Filtration**(الترشيح): separating a solid from liquid using filter

Example: SiO_2

اهم 3 طرق

- **Distillation**(التقطير): vaporizing a liquid from a solid or another liquid and condensing the vapor.
- **Crystallization**(التبلور): forming a crystalline solid decreasing its solubility as a result of cooling the solution, evaporating the solvent, or adding a solvent in which it is less soluble so that solid crystals form.
- **Centrifugation**(الطرد المركزي): removing a solid substance from solution by means of centrifuge.

افهم اخر 3 طرق بس, عشان نادر بيحي عليهم
اسئلة

- Melting point $\longrightarrow$ $\text{NH}_4\text{Cl}_{(s)} < \text{NaCl}_{(s)} < \text{SiO}_{2(s)}$

- Solubility

$\text{NH}_4\text{Cl} \longrightarrow$ soluble

$\text{NaCl} \longrightarrow$ soluble

$\text{SiO}_2 \longrightarrow$ insoluble

All of them appear as white fumes

IN THIS EXPERIMENT : we have a dish and inside it we have ($\text{NH}_4\text{Cl}/\text{NaCl}/\text{SiO}_2$) at the first step we use heat and sublimation to remove NH_4Cl , what stays are NaCl and SiO_2 then we use extraction in addition of water to remove NaCl What remain is SiO_2 and we collect it by filtration.

$$\text{Mass \% of each component} = \frac{\text{mass of component}}{\text{mass of original mixture}} * 100\%$$

$\Sigma \text{ mass \%} = 100\%$! اذا ما كان مجموعهم 100 يكون حلك غلط

Exercise: A mixture weighs 9.8712g and contains 6.3164g of sodium chloride and 3.4779g of silicon dioxide. The remainder of the mixture is ammonium chloride, calculate the percent by weight of each substance in the mixture.

Solution

1) Mass of mixture = 9.8712g.....من السؤال

2) Mass of NaCl = 6.3164gمن السؤال

3) Mass of SiO₂ = 3.4779g.....من السؤال

4) Mass of mixture = mass of NH₄Cl + mass of NaCl + mass of SiO₂

$$9.8712 = \text{mass of NH}_4\text{Cl} + 6.3164 + 3.4779$$

$$\text{Mass of NH}_4\text{Cl} = 0.0769\text{g}$$

5) Percentage of NaCl = $\frac{\text{mass of NaCl}}{\text{mass of mixture}} * 100\% = 63.988\%$

6) Percentage of SiO₂ = $\frac{\text{mass of SiO}_2}{\text{mass of mixture}} * 100\% = 35.2335\%$

7) Percentage of NH₄Cl = $\frac{\text{mass of NH}_4\text{Cl}}{\text{mass of mixture}} * 100\% = 0.7790\%$

QUESTIONS

- 1) A mixture weighs 3.15g and contains only (SiO_2 and NH_4Cl), when this mixture was heated, its mass was decreased to 1.75g. calculate the mass % of SiO_2 AND NH_4Cl in this mixture.

Answer: mass of SiO_2 = 1.75grams because after heating NH_4Cl sublimates, so the mass of NH_4Cl would be $\rightarrow$ mass of mixture = mass of SiO_2 + mass of NH_4Cl

Mass of NH_4Cl = 1.4grams

$$\% \text{SiO}_2 = \frac{1.75}{3.15} * 100\% = 55.5556\%$$

$$\% \text{NH}_4\text{Cl} = \frac{1.4}{3.15} * 100\% = 44.444\%$$

- 2) A mixture containing only NH_4Cl and SiO_2 has a mass of 4.35g, it was heated until vapors were evolved and re-weight to be 2.31g, the mass% of NH_4Cl in this mixture is :

A)33.0

B)67.0

C)46.9

D)53.1

Answer = c

- 3) True or False : NaCl is separated by filtration.

Answer : false, it is separated by extraction

- 4) When we separate NH_4Cl by sublimation is the reaction considered exothermic or endothermic ?

Answer : endothermic

قلنا قبل انو لازم حرارة عشان يتم تفاعل التسامي ل

NH_4Cl

- 5) True or False :

- NH_4Cl is soluble in water but SiO_2 is insoluble
 - you can use sublimation technique to separate solid SiO_2 from solid from solid NH_4Cl
- answer : true/true

EXPERIMENT (3)

LIMITING REACTANT

Objective: to determine the limiting reactant and the percent composition of a salt mixture .

Limiting reactant: is the reactant that is consumed completely first and also has the lowest mole ratio(نسبة) .

- Two factors limit the yield of products in a chemical reaction:
 - 1)the amounts of the reactants
 - 2)the percent yield of the reaction
- Temperature and Pressure can be adjusted to increase the reaction's yield.

How can we determine the limiting reactant and the excess reactant ? the following example will simplify things.



To determine the limiting and excess reactants we divide the number of moles to the coefficient(المعامل)

$$\text{For A} \rightarrow \frac{\text{عدد المولات 5}}{\text{المعامل لـ A 4}} = 1.25$$

$$\text{For B} \rightarrow \frac{8}{7} = 1.142$$

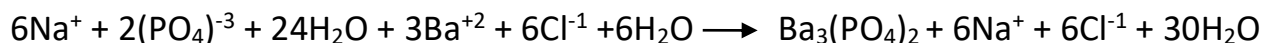
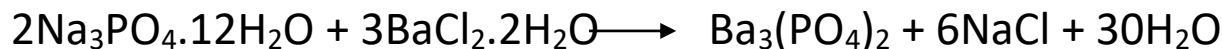
Because B has the lowest mole ratio it will be the limiting reactant

Because A has the highest mole ration it will be the excess reactant

$$\% \text{yield} = \frac{\text{actual yield}}{\text{theoretical yield}} * 100\%$$

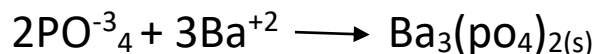
..... مهم

$$\text{Moles} = \frac{\text{mass}}{\text{molar mass}}$$



Example: suppose we only have 700.0grams of $\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}$ (F.W=380.2 g/mol) and 700.0grams of $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$ (FW=244.2 g/mol) dissolve in water and are mixed. How many grams of $\text{Ba}_3(\text{PO}_4)_2$ (FW=602.0g/mol) are then produced?

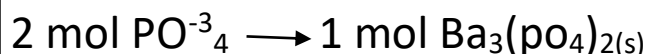
To solve this question we must determine the limiting reactant because our calculations will depend on it(not the excess reactant).



$$\text{Moles of } \text{PO}_4^{-3} = \frac{\text{mass}}{\text{molar mass}} = \frac{700}{380.2} = 1.841 \text{ moles} \rightarrow \text{ratio} = \frac{1.841}{2} = 0.92$$

$$\text{Moles of } \text{Ba}^{+2} = \frac{\text{mass}}{\text{molar mass}} = \frac{700}{244.2} = 2.8665 \text{ moles} \rightarrow \text{ratio} = \frac{2.8665}{3} = 0.95$$

PO_4^{-3} is the limiting reactant because it has the lowest mole ratio.



ضرب تبادلي

$$\text{Ba}_3(\text{po}_4)_{2(s)} \text{ moles} = 0.92 \text{ mol}$$

$$\text{Mole} = \text{mass}/(\text{molar mass}) \Rightarrow \text{mass} = (0.92) \cdot (602) = 554 \text{ gram}$$

Note: $\text{Ba}_3(\text{po}_4)_{2(s)}$ is insoluble in water and in basic medium, it appears as white precipitate, it is soluble only in acidic medium.

$$\% \text{yield} = \frac{\text{actual yield}}{\text{theoretical yield}} * 100\%$$

1) %yield should be less than 100% because :-

A) The reaction may not be completed

B) Loss of sample or product may occur during weighing or filtration or transformation

2) if the %yield is greater than 100% your product is wet (not dried well).

3) the reactant that forms cloudy solution when added is the limiting one.

Addition of 0.5M BaCl_2 solution to supernatant and white ppt formed means that in your unknown Ba^{+2} is the limiting reactant and PO_4^{-3} is the excess.

- Addition of 0.5M Na_3PO_4 solution to supernatant and white ppt formed means that in your unknown PO_4^{-3} is the limiting reactant and Ba^{+2} is excess.

QUESTIONS

1) True or False

- The limiting reactant is the reactant which has the smallest number of moles in the reaction mixture.
- Washing of $\text{Ba}_3(\text{PO}_4)_2(s)$ with acid will decrease the actual yield of precipitate.

Answer: false; because the limiting reactant is the reactant that has the lowest mole ratio.

Answer:false

2) The limiting reactant is the reactant :

- A)that has the smallest number of moles
- B)that has the largest number of moles
- C)that has the smallest mass
- D)that is consumed completely first

Answer:D

3) if the product obtained (limiting reactant experiment) is weighed after drying the percent yield will be Compared to the actual value.

A)do not change B)lower C)higher D)none of these

Answer: C) higher

4) 4.40grams of $\text{Ca}_3(\text{PO}_4)_2$ (MM=310) was reacted with excess HCl according to: $\text{Ca}_3(\text{PO}_4)_2 + 4\text{HCl} \rightarrow \text{Ca}(\text{H}_2\text{PO}_4)_2 + 2\text{CaCl}_2$

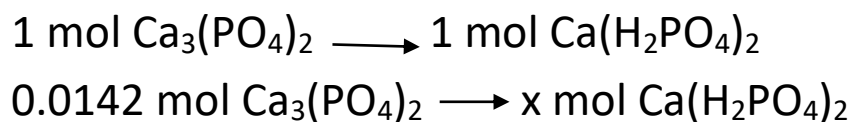
If 2.49gram of $\text{Ca}(\text{H}_2\text{PO}_4)_2$ (MM= 234) was produced, then the percent yield is ?

A)85 B)65 C)75 D)55

Limiting reactant بهيئك سؤال لازم نحدد

HCl is the excess reactant so $\text{Ca}_3(\text{PO}_4)_2$ will be the limiting reactant.

$$\text{Moles } \text{Ca}_3(\text{PO}_4)_2 = \frac{\text{mass}}{\text{molar mass}} = \frac{4.4}{310} = 0.0142 \text{ moles}$$



ضرب تبادلي

$$\text{Moles } \text{Ca}(\text{H}_2\text{PO}_4)_2 = 0.0142 \text{ mol}$$

$$\text{Moles } \text{Ca}(\text{H}_2\text{PO}_4)_2 = \frac{\text{mass}}{\text{molar mass}} \rightarrow \text{mass} = 3.3212 \text{ grams}$$

$$\% \text{yield} = \frac{\text{actual}}{\text{theoretical}} * 100\% = \frac{2.49}{3.3212} * 100\% = 74.9 \text{ answer is C}$$